Corollary

- **Corollary**

Let n be a positive integer. Then

$$\sum_{k=0}^{n} 2^k \binom{n}{k} = 3^n$$

- **Proof**

Using the binomial theorem with $x = 1$ and $y = 2$, we can get

$$3^n = (1+2)^n = \sum_{k=0}^{n} \binom{n}{k} 1^{n-k} 2^k = \sum_{k=0}^{n} \binom{n}{k} 2^k$$

This proves the corollary.